



EDUCATION

BTech - Robotics & Autonomous Systems	Plaksha University, Mohali	9.01/10	2027
12th - CBSE	DAV Public School, Haldia	94.4%	2022
10th - CBSE	DAV Public School, Haldia	94.6%	2020

EXPERIENCE

- Summer Intern** | Robotics Lab, Plaksha June 2025 - Aug 2025
- Developing an **autonomous robot** for crop inspection, utilising sensors and **Machine Learning** for leaf-level disease and pest detection in commercial and horticultural crops.
 - Collaborated on optimising navigation, data analysis, and real-time crop health monitoring to advance **precision farming**.
- Senior Teaching Assistant** | Young Technology Scholar, Plaksha Mar 2025 - Jun 2025
- Fabricated a **strain sensor** from scratch using conductive materials and other chemicals.
 - Built multiple mini-projects to explore sensor applications, and compared **strain vs flex sensors** for performance and responsiveness.
 - Integrated sensor with a robotic arm to demonstrate **prosthetic control**.
- Research Intern** | Center for Thinking, Language & Communication, Plaksha Jun 2024 - Jun 2025
- Developed a research paper on **project-based learning**, which is published by **ASEE** under the supervision of Prof. Prince.
 - Co-authored paper on experiential learning frameworks and project-based approaches to education.
- Summer Intern** | Dixon IoT Lab, Plaksha Jun 2024 - Nov 2024
- Working on advancing solutions for **smart traffic and agricultural management systems** under the mentorship of Prof. Srinivasan.
 - Worked with **IoT hardwares** and different networking protocols like **MQTT, CoAP, AMQP**.

PROJECTS

- Intrusion Detection System** | Prof. Sobti Aug 2025 - Dec 2025
- Developing a **multiclass IDS** using tabular network traffic data, progressing from clustering baselines to **Deep Learning** models by converting tabular features into image-like inputs.
 - Explored various architectures, including **LSTM, CNN, Autoencoders, Knowledge Distillation and Transformer**, achieving best performance with a hybrid **Encoder-Decoder → X-LSTM → CNN** pipeline for spatial-temporal pattern learning.
 - Reached **98.1%** accuracy and F1-score with the final model.
- Crop Health Analysis** | Prof. Siddharth Feb 2025 - May 2025
- Developed a **Deep Learning** model for **multi-crop health classification** using satellite imagery and weather data.
 - Implemented a four-class system (Healthy, Stressed, Diseased, Pest-Infested) for more targeted agricultural insights.
 - Applied **CNN** for spatial-spectral feature extraction and **LSTM** for time-series analysis, comparing ML and DL models on multi-spectral data performance.
 - Reached **80%** accuracy and F1-score with the final model.
- CanSat Satellite Competition** | Prof. Tamaskar Sep 2024 - Feb 2025
- Developing a CanSat satellite designed to reach an **altitude of 1.1 km** above sea level, collecting **telemetry data** such as altitude, temperature, battery voltage, and GPS position.
 - Focusing on creating the **ground station interface** to display data on a UI and conducting unit testing for all integrated sensors.
 - Top 10** in India and **Top 45** in the world with a score of **92.4%**.

POSITIONS OF RESPONSIBILITY

- Class Representative, RAS** | Plaksha University
- Liaised between students and faculty, addressing concerns and ensuring effective communication.
- Environmental Captain** | DAV Public School
- Led green initiatives and environmental campaigns within school.

SKILLS

- Programming Languages:** C++, C, Python, MATLAB
- Web Development:** HTML, CSS, JS
- IoT:** Raspberry Pi, Networking Protocols, ESP32
- Technologies:** Machine Learning, Deep Learning, MySQL
- Operating Systems:** Linux, ROS

ACHIEVEMENTS

- Runners-up, Unstop Hackathon** – Developed a soil health monitoring system using IoT and ML for real-time agricultural insights (received incubation opportunity from IIT Ropar).
- Certificate of Distinction**- Australian National Chemistry Quiz, Royal Australian Chemical Institute (International Event)
- Certificate of Excellence**- All India Essay Competition 2018, UN Information Centre (Participated by schools across India)